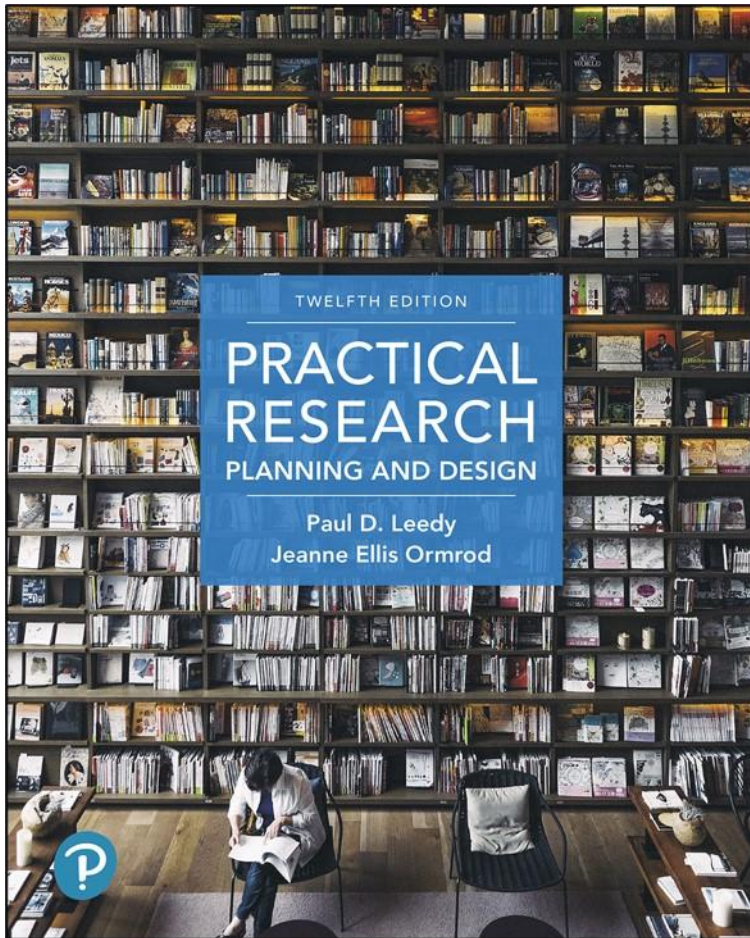


Practical Research: Planning and Design

Twelfth Edition



Chapter 7

Experimental, Quasi-
Experimental, and Ex Post Facto
Designs

Experimental Studies

- Used to test for cause-and-effect relationships
- Many possible factors might cause or influence a particular condition or phenomenon
- The researcher controls for all influential factors except those whose possible effects are the focus of investigation

Variables

- Variable: quality or characteristic that has two or more values
 - Independent Variable
 - the researcher studies as a possible cause of something else
 - the researcher directly manipulates
 - Dependent Variable
 - potentially influenced by (depends on) the independent variable

Internal Validity

- The extent to which the design of a research study and the data it yields allow the researcher to draw accurate conclusions about cause-and-effect and other relationships
 - Experimental designs must be internally valid, or the results are not interpretable

Confounding Variables (1 of 2)

- Factors that might account for differences that are not attributable to the particular treatment or intervention being studied

Confounding Variables (2 of 2)

- Example:
 - Two classes use different methods of teaching science
 - Teaching method = independent variable
 - Two different teachers = confounding variable
 - Teachers of different sex = another confounding variable
 - Time of day the class taught = another confounding variable

Strategies for Controlling Confounding Variables (1 of 2)

1. Keep some things constant
 - Example: Heterogeneous sample
2. Include a control group
3. Conduct a double-blind experiment
 - participants and individuals administering varying interventions or control conditions are equally “in the dark”
4. Randomly assign people to groups

Strategies for Controlling Confounding Variables (2 of 2)

5. Use one or more pretests to assess equivalence before the treatment(s)
 - matched pairs
 6. Expose participants to all experimental treatments
 - within-subjects or repeated-measures variablea
 7. Statistically control for confounding variables
 - partial correlation, ANCOVA, SEM
- Remember, a carefully controlled experimental design is the only approach that allows you to draw firm conclusions about cause and effect.

Experimental Designs

- “True” experimental designs have a truly independent variable, manipulated by the researcher
 - Experimental designs offer control and internal validity
- Variations:
 - Pre-experimental designs
 - Quasi-experimental designs
 - Ex post facto designs
 - Factorial designs

Pre-Experimental Designs (1 of 2)

- Differ from experimental designs in that:
 - the independent variable doesn't vary
 - experimental and control groups are not composed of equivalent or randomly selected individuals
- Not possible to show cause-and-effect relationships
- Should be followed by more controlled experiments

Pre-Experimental Designs (2 of 2)

- One-Shot Experimental Case Study
 - Group 1 Tx → Obs
- One-Group Pretest-Posttest Design
 - Group 1 Obs → Tx → Obs
- Static Group Comparison
 - Group 1 Tx → Obs
 - Group 1 --- Obs

True Experimental Designs (1 of 2)

- Pretest-Posttest Control Group Design (random assignment)

Group		Time →		
Random Assignment	Group 1	Obs	Tx	Obs
	Group 2	Obs	—	Obs

- Solomon Four-Group Design (enhances external validity)

Group		Time →		
Random Assignment	Group 1	Obs	Tx	Obs
	Group 2	Obs	—	Obs
	Group 3	—	Tx	Obs
	Group 4	—	—	Obs

True Experimental Designs (2 of 2)

- Posttest Only Control Group Design

	Group	Time →	
Random Assignment	Group 1	Tx	Obs
	Group 2	—	Obs

- Within Subjects Design

Group	Time →	
Group 1	Tx _a	Obs _a
	Tx _b	Obs _b

Quasi-Experimental Design (1 of 3)

- Randomness is not possible or practical
 - can't control for all confounding variables
- Nonrandomized Control-Group Pretest-Posttest Design

Group	Time →		
Group 1	Obs	Tx	Obs
Group 2	Obs	—	Obs

- Simple Time-Series Design

Group	Time →								
Group 1	Obs	Obs	Obs	Obs	Tx	Obs	Obs	Obs	Obs

Quasi-Experimental Design (2 of 3)

- Control-Group Time-Series Design

Group	Time →								
Group 1	Obs	Obs	Obs	Obs	Tx	Obs	Obs	Obs	Obs
Group 2	Obs	Obs	Obs	Obs	—	Obs	Obs	Obs	Obs

- Reversal Time-Series Design

Group	Time →							
Group 1	Tx	Obs	—	Obs	Tx	Obs	—	Obs

Quasi-Experimental Design (3 of 3)

- Alternating-Treatments Design

Group	Time →													
Group 1	Tx _a	Obs	—	Obs	Tx _b	Obs	—	Obs	Tx _a	Obs	—	Obs	Tx _b	Obs

- Multiple Baseline Design

Group	Time →					
Group 1	Baseline →			Treatment →		
	—	Obs		Tx	Obs	
Group 2	Baseline →				Treatment →	
	—	Obs		—	Obs	

Single-Subject Designs

- Reversal time-series, alternating treatments, and multiple-baseline designs might also be used with either a single individual or a single intact group.
- This approach is known as single-case intervention research.

Ex Post Facto Design (1 of 3)

- Researcher identifies events that have already occurred or conditions that are already present and collects data to investigate a possible relationship between these factors and subsequent characteristics or behaviors
- Often confused with correlational or experimental designs

Ex Post Facto Design (2 of 3)

- involves looking at existing circumstances (like correlational research)
- has clearly identifiable independent and dependent variables (like experimental research)
- involves no direct manipulation of the independent variable – the presumed “cause” has already occurred (unlike experimental research)

Ex Post Facto Design (3 of 3)

- Simple Ex Post Facto Design

Group	Time →	
	Prior event (s)	Investigation Period
Group 1	Exp	Obs
Group 2	—	Obs

Factorial Design (1 of 2)

- Two or more independent variables in a single study
- Two-Factor Experimental Design

Group		Time →		
Random Assignment		Treatments related to the two variables may occur simultaneously or sequentially		
		Treatment related Variable 1	Treatment related Variable 2	
	Group 1	Tx_1	Tx_2	Obs
	Group 2	Tx_1	—	Obs
	Group 3	—	Tx_2	Obs
	Group 4	—	—	Obs

Factorial Design (2 of 2)

- Combined experimental and ex post facto design

Group	Time →				
	Prior event(s)	Investigation Period →			
Group 1	Exp _a	Random Assignment	Group 1a	Tx _a	Obs
			Group 1b	Tx _b	Obs
Group 2	Exp _b	Random Assignment	Group 2a	Tx _a	Obs
			Group 2b	Tx _b	Obs

Meta-Analysis

- Statistical technique: Analysis of the analyses
- Researcher combines results of many experimental and/or ex post facto studies to determine whether they lead to consistent conclusions

Conducting Experiments on the Internet

- Some websites are set up to assist you
- Keep ethics in mind
- Remember your sample will probably not be representative

Acknowledging Bias

- Occurs even in the most controlled designs
 - variables may be ignored
 - sample may not be representative
 - tasks or setting may be artificial
- Identify it, acknowledge it, and report it

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